Lecture 8: Planetary Interiors
Structure, Composition, & Dynamics

Planetary Systems

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Earth's interior cross-section. Credit: NASA/JPL-Caltech

Learning Objectives

By the end of this lecture, you will be able to:

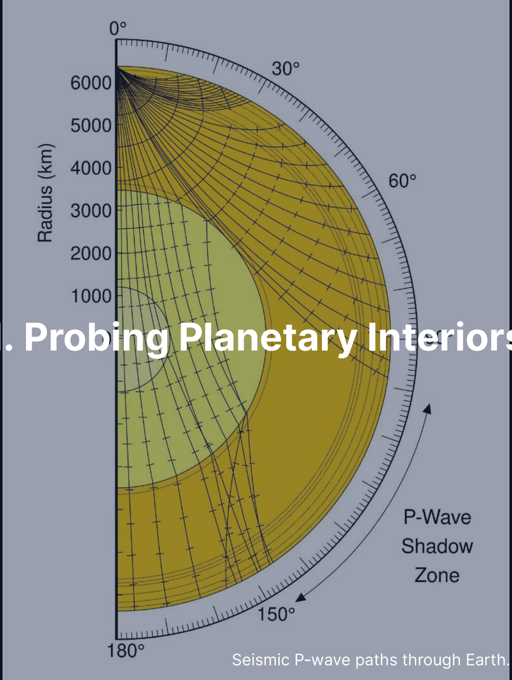
- Explain how seismology, gravity, and the moment of inertia probe planetary interiors
- Derive C/MR^2 for uniform and two-layer bodies
- Apply hydrostatic equilibrium and equations of state at planetary conditions
- Compare the internal structures of terrestrial planets, giant planets, and icy moons
- Describe recent results from InSight, Juno, and Europa Clipper

Recap: Lecture 7

- Impact cratering: morphology, scaling laws, crater counting
- Volcanism: effusive vs. explosive, shield volcanoes, flood basalts
- Tectonics: extensional, compressional, strike-slip faulting
- Erosion and weathering: wind, water, ice, space weathering
- Cryovolcanism on icy bodies (Enceladus, Triton)

Today: What lies *beneath* the surface? How do we know, and what does it tell us?

1. Probing Planetary Interiors



Seismic P-wave paths through Earth. Credit: Wikimedia Commons, public domain

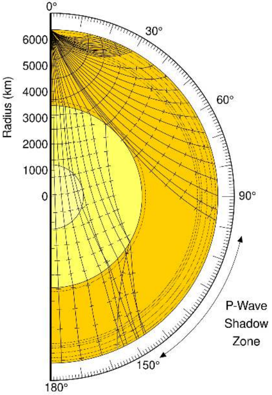
Three Ways to See Inside a Planet

We cannot directly sample deep interiors. Three complementary tools:

1. **Seismology:** wave propagation through layers of different density and rigidity
2. **Gravity field measurements:** mass distribution from spacecraft tracking
3. **Moment of inertia:** degree of central mass concentration

For Earth: all three are available at high precision.
For other planets: gravity and C/MR^2 are often all we have.

Seismic Waves: P-Waves and S-Waves



Credit: Wikimedia Commons, public domain

P-waves (primary, compressional):

- Travel through solids *and* liquids
- $v_P = \sqrt{(K + \frac{4}{3}\mu)/\rho}$

S-waves (secondary, shear):

- Travel through solids *only*
- $v_S = \sqrt{\mu/\rho}$
- $\mu = 0$ in a liquid $\Rightarrow$ S-waves **cannot propagate**

The Shadow Zone: Proof of a Liquid Core

- P-waves are refracted by the liquid outer core, creating a **P-wave shadow zone** between ~104° and ~140° from the epicentre
- S-waves are **entirely absorbed** by the liquid core: shadow beyond 104°
- Discovered by Oldham (1906) and Gutenberg (1914)

The absence of S-waves through the outer core is the definitive evidence that it is liquid.

Gravity Field Measurements

- A perfectly spherical, homogeneous body produces a pure $1/r^2$ field
- Real planets are oblate due to rotation $\Rightarrow$ **gravitational quadrupole** J_2
- J_2 depends on mass distribution between equator and poles
- Higher harmonics reveal lateral density variations
- Measured via spacecraft radio tracking (Doppler shifts)
- Examples: GRAIL (Moon), Juno (Jupiter), MRO (Mars)

Precision: Juno measures Jupiter's gravity harmonics to parts in 10^9.

The Moment of Inertia Factor: C/MR^2

The **most diagnostic single number** for internal mass distribution:

Body	C/MR^2	Interpretation
Uniform sphere	0.400	Homogeneous, undifferentiated
Moon	0.393	Weakly differentiated, small core
Mars	0.364	Moderately differentiated
Mercury	0.346	Large dense core
Earth	0.331	Strongly differentiated, iron core

Source: de Pater & Lissauer (2010), Table 6.1

Determined from J_2 + spin-axis precession rate (or libration amplitude for tidally locked bodies), combined via the **Darwin–Radau relation**.

Measuring C/MR^2 in Practice

Two observables needed:

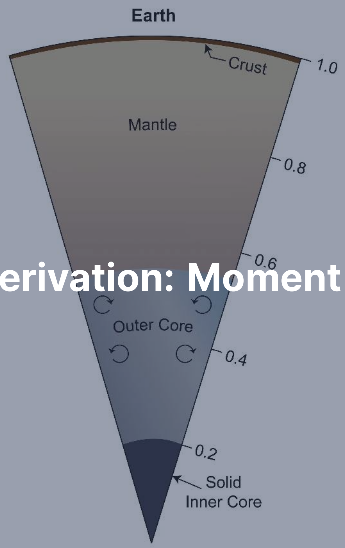
1. **Gravitational oblateness** J_2 from spacecraft tracking
2. **Precession rate** of the spin axis (or forced libration for synchronous rotators)

These combine via the **Darwin–Radau relation**:

$$\frac{C}{MR^2} = \frac{2}{3}\left[1 - \frac{2}{5}\sqrt{\frac{4 - k_f}{1 + k_f}}\right]$$

where k_f is the fluid Love number (a measure of the body's response to rotation).

Result: a single number that immediately distinguishes a rubble pile (0.4) from a differentiated body like Earth (0.33).



2. Blackboard Derivation: Moment of Inertia Factor

Earth’s interior. Credit: NASA/JPL-Caltech

Deriving C/MR^2

Goal: Derive the moment of inertia factor for (a) a uniform sphere and (b) a two-layer differentiated body. Show how the measured value constrains core size.

Starting point: For a spherically symmetric body, the moment of inertia about the rotation axis is:

$$C = \frac{8\pi}{3} \int_0^R \rho(r)\, r^4 \,\mathrm{d}r$$

This comes from summing $\frac{2}{3} r^2 \,\mathrm{d}m$ over thin spherical shells.

Step 1: Uniform Sphere

Set ρ = const:

$$C = \frac{8\pi}{3}\rho \int_0^R r^4 \, \mathrm{d}r = \frac{8\pi}{3}\rho \frac{R^5}{5} = \frac{8\pi}{15}\rho R^5$$

Total mass: $M = \frac{4}{3}\pi R^3 \rho$

$$\frac{C}{MR^2} = \frac{8\pi\rho R^5/15}{(4\pi\rho R^3/3)\,R^2} = \frac{2}{5} = 0.400$$

This is the **maximum** for any body with $\rho(r)$ that does not increase outward.

Step 2: Two-Layer Body

Dense core (ρ_c, radius R_c) + lighter mantle (ρ_m, radius R):

$$C = \frac{8\pi}{15}\left[(\rho_c - \rho_m)\,R_c^5 + \rho_m\,R^5\right]$$

$$M = \frac{4}{3}\pi\left[(\rho_c - \rho_m)\,R_c^3 + \rho_m\,R^3\right]$$

Define $x \equiv R_c/R$ and $f \equiv \rho_c/\rho_m$:

$$\frac{C}{MR^2} = \frac{2}{5} \cdot \frac{(f-1)\,x^5 + 1}{(f-1)\,x^3 + 1}$$

Since $x^5 < x^3$ for $0 < x < 1$, the ratio is < 1 when $f > 1$.
More mass in the centre ⇒ lower C/MR^2.

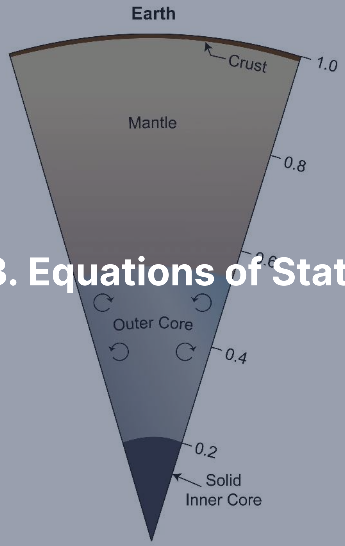
Worked Example: Earth

Parameter	Symbol	Value
Core radius fraction	$x = R_c/R$	$3480/6371 = 0.546$
Density ratio	$f = \rho_c/\rho_m$	$11{,}000/4{,}400 \approx 2.5$

$$\frac{C}{MR^2} = \frac{2}{5} \times \frac{1.5 \times 0.049 + 1}{1.5 \times 0.163 + 1} = 0.4 \times \frac{1.073}{1.244} \approx 0.345$$

Measured value: $C/MR^2 = 0.3307$

Remaining difference: continuous density increase within each layer, especially the ~30% jump from outer to inner core.



3. Equations of State

Earth's interior. Credit: NASA/JPL-Caltech

Hydrostatic Equilibrium in Spherical Symmetry

Pressure at each depth supports the weight of overlying material:

$$\frac{\mathrm{d}P}{\mathrm{d}r} = -\rho(r)\, g(r) = -\frac{G\, m(r)\, \rho(r)}{r^2}$$

where $m(r) = 4\pi \int_0^r \rho(r')\, r'^2\, \mathrm{d}r'$ is the mass enclosed within radius r.

Combined with an equation of state $P = P(\rho, T)$, this can be integrated inward from the surface to determine the full pressure–density profile.

Central Pressure Estimate

For uniform density $\bar{\rho} = 3M/(4\pi R^3)$:

$$P_c \approx \frac{3\,G\,M^2}{8\pi\,R^4}$$

Body	Estimated P_c	Actual P_c
Earth	~170 GPa	~360 GPa
Mars	~25 GPa	~40 GPa
Jupiter	~700 GPa	~4000 GPa

The uniform-density estimate is always **too low** because real bodies have higher density at the centre.

Birch's Law

Birch (1952): empirical linear relationship between compressional wave velocity v_P and density ρ for silicates and oxides:

$$v_P = a(\bar{M}) + b\,\rho$$

where $\bar{M}$ is the mean atomic weight and a, b are empirical constants.

- Connects seismology directly to density
- Holds remarkably well for mantle minerals at high pressure
- Breaks down at phase transitions and compositional boundaries

Birch's law is a cornerstone of mineral physics, linking lab measurements, seismic data, and interior models.

Adams–Williamson Equation

The **seismic parameter**: $\phi \equiv K_S/\rho = v_P^2 - \frac{4}{3}v_S^2$

Combining with hydrostatic equilibrium:

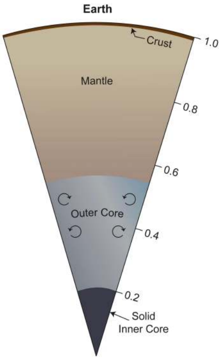
$$\frac{\mathrm{d}\rho}{\mathrm{d}r} = -\frac{\rho(r)\,g(r)}{\phi(r)}$$

- Determines density profile from measured seismic wave speeds
- Valid where composition is approximately uniform
- Breaks down at phase transitions and compositional boundaries
- Foundation of the PREM density model

4. Earth’s Interior

Earth’s interior. Credit: NASA/JPL-Caltech

Earth's Internal Structure



Credit: NASA/JPL-Caltech

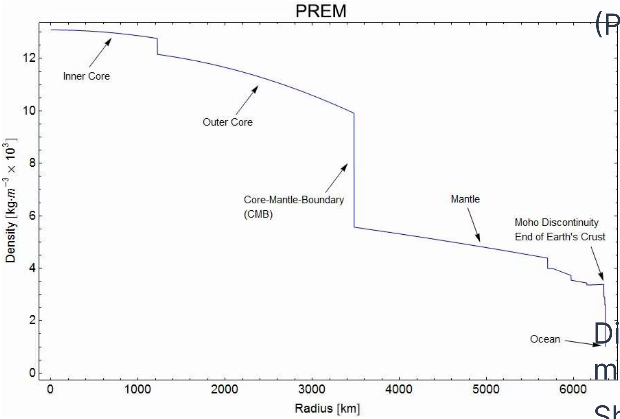
Four major layers:

1. **Crust** (0–5–70 km)
2. **Mantle** (Moho–2891 km)
3. **Outer core** (2891–5150 km): liquid
4. **Inner core** (5150–6371 km): solid

The inner core is solid **because of pressure**: at 330–360 GPa, iron's melting point exceeds the core's 5000–6000 K.

Mean density	5515 kg m^{-3}
C/MR^2	0.3307
Surface temp	~288 K
Central temp	~5000–6000 K
Central pressure	~360 GPa

The PREM Density Profile



Credit: Dziewonski & Anderson (1981)

Preliminary Reference Earth Model (PREM):

- Upper mantle: ~3300 kg m^{-3}
- Lower mantle base: ~5500 kg m^{-3}
- Outer core: ~9900–12,200 kg m^{-3}
- Inner core: ~13,000 kg m^{-3}

Discontinuities at 410 and 660 km: mineral phase transitions.

Sharp jump at CMB (~3480 km): silicate → iron alloy.

The Crust

Oceanic crust:

- 5–10 km thick
- Basaltic, $\rho \approx 3000\ \mathrm{kg\,m^{-3}}$
- Young: <200 Myr (recycled by subduction)

Continental crust:

- 30–70 km thick
- Granitic, $\rho \approx 2700\ \mathrm{kg\,m^{-3}}$
- Old: up to ~4 Gyr

The crust is <1% of Earth's mass, but contains a large fraction of incompatible and heat-producing elements (U, Th, K).

Base of the crust = the **Mohorovičić discontinuity** (Moho), discovered 1909.

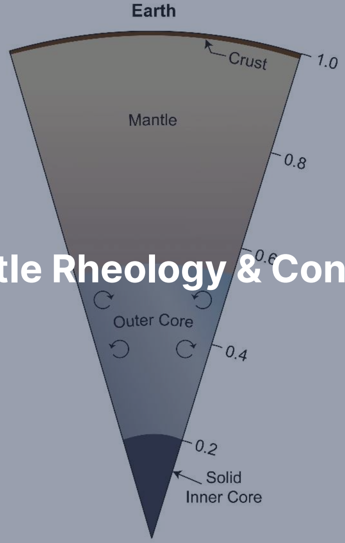
Mantle and Core: Key Numbers

Layer	Depth (km)	ρ (kg m^{-3})	Composition
Upper mantle	0–410	3300–3700	Olivine, pyroxene
Transition zone	410–660	3700–4400	Wadsleyite, ringwoodite
Lower mantle	660–2891	4400–5500	Bridgmanite, ferropericlase
Outer core	2891–5150	9900–12,200	Liquid Fe–Ni + light elements
Inner core	5150–6371	~13,000	Solid Fe–Ni

Source: PREM (Dziewonski & Anderson 1981); Stixrude & Lithgow-Bertelloni (2005)

- Outer core is liquid: no S-waves, drives the geodynamo
- Inner core grows at ~0.5 mm yr^{-1} as Earth cools
- Latent heat from inner core crystallisation helps power the dynamo (Lecture 4)

5. Mantle Rheology & Convection



Earth's interior. Credit: NASA/JPL-Caltech

Solid-State Creep

The mantle is **solid** (transmits S-waves), yet it flows over geological timescales.

- At $T > 0.5\,T_{\mathrm{melt}}$: atoms migrate along grain boundaries and through the lattice
- **Diffusion creep:** viscosity depends on grain size, dominant in lower mantle
- **Dislocation creep:** viscosity depends on stress, dominant in upper mantle

Material	Viscosity (Pa s)
Water	10^{-3}
Glacial ice	10^{12}
Upper mantle	10^{20}–10^{21}
Lower mantle	10^{22}–10^{23}

The Rayleigh Number

Whether the mantle convects is governed by:

$$\mathrm{Ra} = \frac{\alpha\, \rho\, g\, \Delta T\, d^3}{\kappa\, \eta}$$

Parameter	Symbol	Earth value
Thermal expansivity	α	3×10^{-5} K^{-1}
Density	ρ	4000 kg m^{-3}
Gravity	g	10 m s^{-2}
Temperature contrast	ΔT	2500 K
Mantle thickness	d	2.9×10^{6} m
Thermal diffusivity	κ	10^{-6} m^2 s^{-1}
Viscosity	η	10^{21} Pa s

$\mathrm{Ra} \sim 10^7$–$10^8 \gg \mathrm{Ra}_c \approx 10^3 \Rightarrow$ **vigorous convection.**

Convection Patterns: An Open Debate

Whole-mantle convection:

- Material circulates from surface to CMB
- Slabs penetrate through 660 km
- Plumes rise from CMB to surface
- Supported by seismic tomography of some subducting slabs

Layered convection:

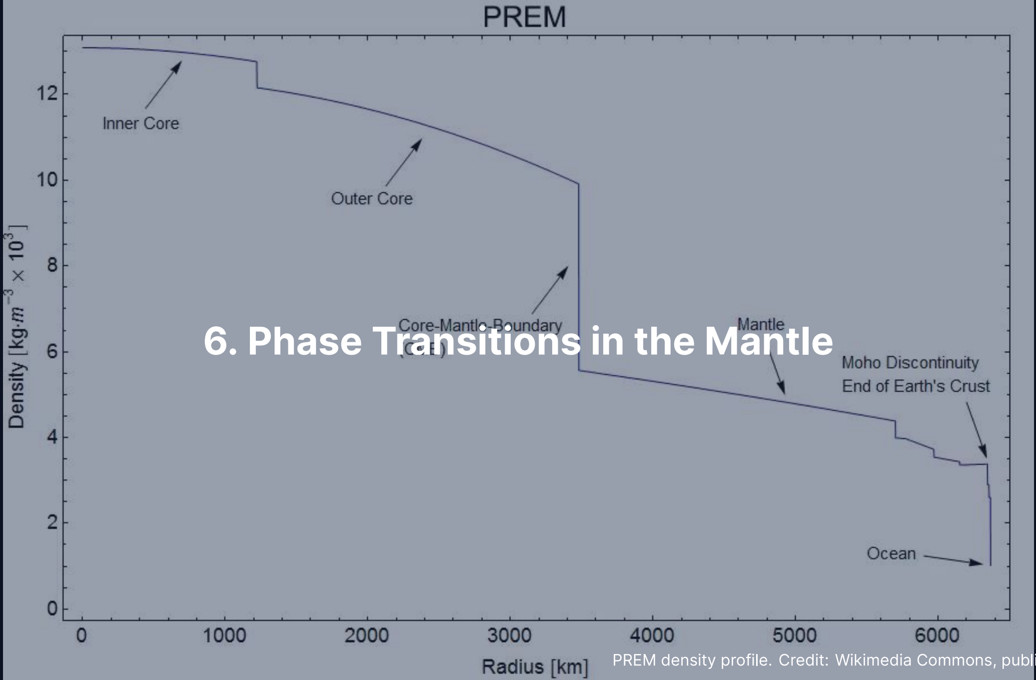
- 660 km discontinuity impedes flow
- Separate upper and lower cells
- Some slabs stagnate at 660 km
- Different chemical reservoirs persist

Reality is likely a **mixed regime**: some slabs penetrate, others stagnate. The 660 km boundary is a partial barrier, not an absolute one.

Break

Earthrise, Apollo 8 (1968). Credit: NASA

6. Phase Transitions in the Mantle



PREM density profile. Credit: Wikimedia Commons, public domain

Mineral Phase Transitions with Depth

Depth (km)	Transition	$\Delta\rho$
410	Olivine → wadsleyite	~5%
520	Wadsleyite → ringwoodite	~2%
660	Ringwoodite → bridgmanite + ferropericlase	~8%
~2700	Bridgmanite → post-perovskite	~1–2%

- Same chemical composition ($(\mathrm{Mg,Fe})_2\mathrm{SiO}_4$), different crystal structures
- Driven by pressure: atoms rearrange into denser packing
- These transitions produce the **seismic discontinuities** in the PREM model

The 660 km Discontinuity

The most important internal boundary within the mantle:

- Ringwoodite breaks down into bridgmanite + ferropericlase
- Largest density jump (~8%) within the mantle
- **Negative Clapeyron slope:** boundary pressure *decreases* with increasing temperature
- A descending hot slab shifts the boundary *deeper*, creating buoyancy resistance
- Partially impedes convective flow: key to the layered vs. whole-mantle debate

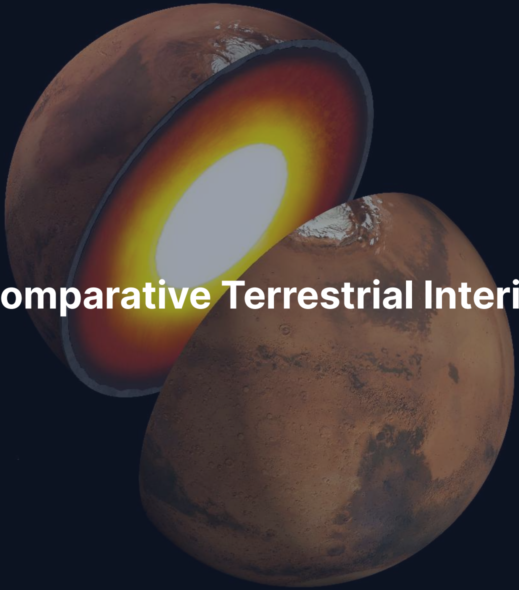
The sign of the Clapeyron slope determines whether a phase transition promotes or resists convective penetration.

Post-Perovskite and the D” Layer

- Discovered in lab experiments in 2004 (Murakami et al.)
- Bridgmanite → post-perovskite at ~125 GPa, ~2500 K
- Occurs ~200 km above the CMB (the **D” layer**)
- May explain:
  - Complex seismic velocity anomalies in D”
  - Lateral variations in the lowermost mantle
  - Anisotropy observed near the CMB

Not unique to Earth: any sufficiently large rocky body will experience these transitions at appropriate pressures.
Mars likely has an olivine–wadsleyite transition; the Moon is too small.

7. Comparative Terrestrial Interiors



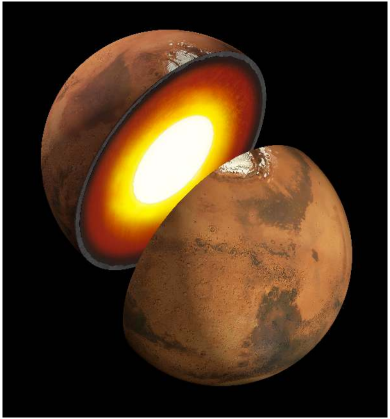
Mars interior from InSight. Credit: NASA/JPL-Caltech

The Moon

- C/MR^2 = 0.393: nearly undifferentiated (close to 0.4)
- $\bar{\rho}$ = 3346 kg m^{-3}
- Crust: ~30–40 km (anorthosite from magma ocean flotation)
- Silicate mantle to ~1400 km depth
- Small, partially molten core: R_c ~ 330–380 km (~20% of lunar radius)

- Apollo seismometers (1969–1977): first extraterrestrial seismology
- Near-uniform structure + low mean density consistent with formation from silicate mantle material (giant impact origin, Lecture 2)

Mars: Revealed by InSight



Credit: NASA/JPL-Caltech

NASA InSight (2018–2022), SEIS seismometer:

- **Crust:** 24–72 km thick
- **Mantle:** to ~1560 km depth
  Olivine–wadsleyite transition at ~1100 km
- **Liquid core:** $R_c \sim 1830$ km (~54% of R)
- Core density ~6000 kg m^{-3} (Earth: ~10,900)
- Rich in sulphur and light elements

No active dynamo today: core not convecting vigorously enough (Lecture 4).

Mercury: An Iron World

- $C/MR^2 = 0.346$, $\bar{\rho} = 5427$ kg m^{-3}
- Highest uncompressed density of any terrestrial planet
- **Enormous iron core:** $R_c \sim 2020$ km, ~85% of the planet's radius
- Core contains ~65% of Mercury's total mass
- Thin silicate mantle (~400 km) + crust (~30–50 km)
- Weak but global magnetic field ⇒ partly liquid, convecting core

MESSENGER spacecraft (2011–2015) + Earth-based radar libration measurements provided these constraints.

Open question: Why such a large core fraction? (Lecture 10)

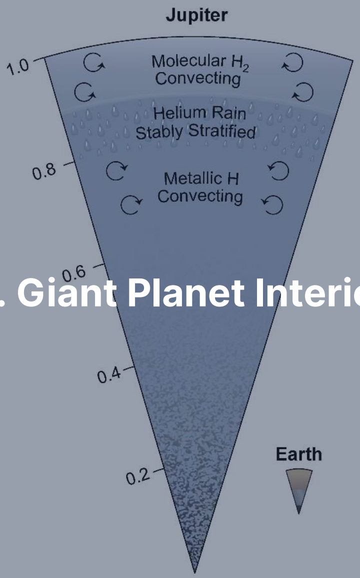
Comparing Terrestrial Body Interiors

Body	$\bar{\rho}$ **(kg m^{-3})**	C/MR^2	R_c/R	**Dynamo?**
Moon	3346	0.393	~0.20	No (ancient)
Mars	3934	0.364	~0.54	No (ancient)
Venus	5243	~0.33	~0.55	No
Earth	5515	0.331	0.55	Yes
Mercury	5427	0.346	~0.85	Weak

Trends:

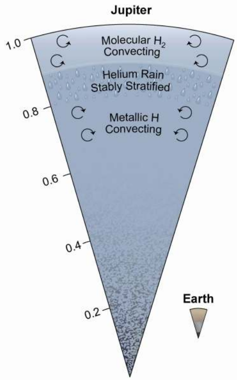
- Higher mean density ⇒ larger iron core fraction (generally)
- Lower C/MR^2 ⇒ stronger differentiation
- Active dynamo requires liquid conducting region + vigorous convection

8. Giant Planet Interiors



Jupiter's interior. Credit: NASA/JPL-Caltech

Jupiter: Metallic Hydrogen and a Dilute Core



Credit: NASA/JPL-Caltech

$M = 318\,M_\oplus$, $R = 11.2\,R_\oplus$

- **Molecular** $\mathrm{H_2}$ **envelope** (outer ~15%): gas → liquid
- **Metallic hydrogen** (to ~80% of R): P > 1–4 Mbar, electrons delocalised, conducts electricity
- **Dilute core:** heavy elements spread over inner ~30–50% of radius
- Central: ~4000 GPa, ~20,000 K

Radiates ~1.7× absorbed solar flux (Kelvin–Helmholtz contraction).

Saturn: Helium Rain

$M = 95\,M_\oplus$, $R = 9.4\,R_\oplus$

- Same overall structure as Jupiter, but lower pressures throughout
- Metallic hydrogen transition occurs deeper
- Larger proportional heavy-element enrichment
- **Helium rain:** as Saturn cools, He becomes immiscible in metallic H
  - Helium droplets settle toward the centre
  - Releases gravitational potential energy
  - Explains Saturn's **excess luminosity**
- Radiates ~2.0× absorbed solar flux (more than contraction alone predicts)

Helium rain is a phase separation driven by cooling, analogous to oil and water unmixing.

Uranus & Neptune: Ice Giant Interiors

Canonical three-layer model:

1. **H/He envelope** (outer ~20% by mass): molecular gas → dense fluid
2. **Ice/fluid layer** (~60–70% by mass): H_2O, NH_3, CH_4
   At 10–500 GPa, 2000–7000 K ⇒ **superionic state**
   (O lattice fixed, H atoms diffuse freely, conducts like a metal)
3. **Rocky core** (~10–20% by mass): silicates + iron

Despite the name "ice," the material is a hot, dense, electrically conducting fluid.

Ice Giant Magnetic Fields: Unusual Geometry

Jupiter/Saturn:

- Roughly dipolar
- Aligned with rotation axis
- Generated in deep metallic H shell

Uranus/Neptune:

- Strongly tilted (59°/47°)
- Offset from centre
- Strongly non-dipolar

The ice giant dynamos likely operate in the thin, shallow conducting ice layer rather than a deep shell, producing complex field geometries.

Uranus puzzle: essentially no measurable internal heat excess, while Neptune emits ~2.6× absorbed solar flux. Possible cause: compositional gradients inhibiting convection in Uranus.

9. Icy Moon Interiors & Ocean Worlds

Europa’s subsurface ocean. Credit: NASA/JPL-Caltech

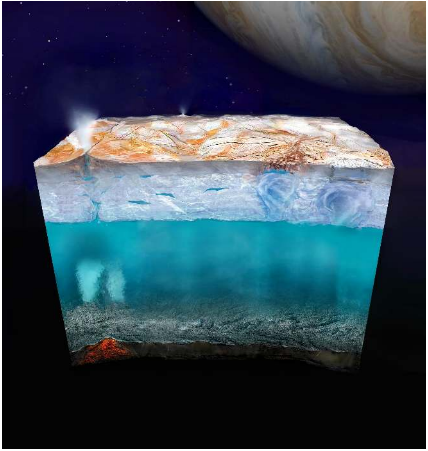
Ocean Worlds: Liquid Water Beneath Ice

Several outer solar system moons harbour **subsurface oceans**:

- Maintained by **tidal heating** (orbital resonances) and radiogenic heat
- Insulating ice shell acts as a thermal blanket
- Prime targets for astrobiology: liquid water + chemical energy + organics

Moon	R **(km)**	ρ **(kg m^{-3})**	**Ocean evidence**
Europa	1561	3013	Magnetic induction, surface geology
Enceladus	252	1609	Geysers, libration
Titan	2575	1882	Non-synchronous rotation
Ganymede	2634	1936	Magnetic field oscillations

Europa: A Habitable Ocean?



Credit: NASA/JPL-Caltech

- Ice shell: ~15–25 km thick
- Global ocean: ~60–150 km deep
- Contains 2–3× the volume of Earth's oceans
- Rocky silicate mantle beneath

Evidence:

- Galileo: induced magnetic field (conducting layer)
- Chaos terrain: disrupted ice, like sea ice break-up
- Tidal heating from Laplace resonance (Io–Europa–Ganymede)

Enceladus: Direct Ocean Sampling

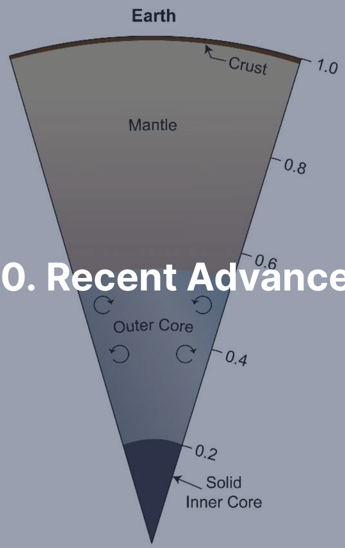
- R = 252 km, remarkably small for an active world
- Cassini flew through **geysers** erupting from "tiger stripe" fractures near the south pole
- Plume composition:
  - Water vapour and ice particles
  - Salts ($NaCl$, $NaHCO_3$)
  - Silica nanoparticles ⇒ hydrothermal activity (>90°C on seafloor)
  - Molecular H_2: potential energy source for chemosynthetic life
- Global subsurface ocean confirmed by libration measurements
- Maintained by tidal heating from resonance with Dione

Titan: Ocean Beneath a Thick Shell

- R = 2575 km, ρ = 1882 kg m^{-3}
- ~50:50 rock–ice composition
- **Subsurface water–ammonia ocean** beneath ~50–100 km ice shell
- Evidence: Cassini radar detected slight non-synchronous rotation (decoupled shell floating on liquid)
- Dragonfly rotorcraft mission: arrival in the 2030s
- Will explore surface chemistry and prebiotic conditions

Ocean worlds demonstrate that the classical habitable zone is far too narrow. Tidal and radiogenic heating can maintain liquid water well beyond the snow line.

10. Recent Advances



Earth’s interior. Credit: NASA/JPL-Caltech

The Mass–Radius Diagram

A single diagnostic plot for solar system planets and exoplanets.

- Rocky planets: $R \propto M^{1/3.7}$ (bulk composition roughly similar)
- Ice/water worlds: shifted to larger radius at given mass
- Gas-envelope worlds: even larger R; Neptunes and sub-Neptunes
- Gas giants: maximum R near 1 Jupiter radius, nearly independent of mass
- Exoplanet surveys: thousands of measurements populate the diagram
- **Radius valley** at 1.5–2 $R_\oplus$: gap between super-Earths and mini-Neptunes

Source: Zeng et al. (2019); Fulton et al. (2017)

Inferring Composition from Mass and Radius

Density alone is highly degenerate; mass and radius together constrain composition.

- Earth (1 $M_\oplus$, 1 $R_\oplus$): average ρ = 5515 kg/m^3, rock + Fe core
- A pure "iron planet" at 1 $M_\oplus$ would have $R \approx 0.75\,R_\oplus$
- A pure "water world" at 1 $M_\oplus$ would have $R \approx 1.3\,R_\oplus$
- Intermediate compositions trace curves between these extremes
- With measurement errors, compositions are often ambiguous (rock vs. mixed)
- Tidal Love number k_2 can break degeneracies: probes internal rigidity

Source: Valencia et al. (2007); Unterborn & Panero (2019)

InSight: First Seismology on Mars Since Apollo

NASA InSight (2018–2022) delivered landmark results:

- Detected hundreds of marsquakes with the SEIS instrument
- **Liquid iron-alloy core:** R_c ~ 1830 km (Stähler et al. 2021)
- **Thick lithosphere:** ~500 km, with transition zone at ~1100 km (Khan et al. 2021)
- **Crust:** 24–72 km thick (thin in north, thick in south)
- Confirmed that Mars is differentiated but has a proportionally larger, less dense core than expected from simple two-layer models

First seismological constraints on another planet's interior since Apollo (1969–1977).

Reading a Marsquake Waveform

InSight’s SEIS instrument revealed Mars’ interior through wave-by-wave analysis.

- **P-wave arrival:** first; travels ~8 km/s through the mantle
- **S-wave arrival:** second, ~4 km/s; absent below the CMB (liquid core)
- **S-wave to P-wave delay:** measures source distance (~8 s → ~1000 km epicentre)
- Core reflections (ScS): bounce off the CMB; constrain core radius
- Mantle surface waves: constrain crustal thickness and attenuation
- Largest marsquake: **S1222a, May 2022, M 4.7**; clearly identified core phases

Every quake is a probe. InSight’s ~1300 events together built a radial model of Mars’ interior.

Juno: Jupiter's Dilute Core

thermodynamical properties of Jupiter for models with/without a small pure heavy-element core. This will allow us to understand how the requirement for a small compact core is compensated, and whether this is possible. If a detailed analysis on the requirement for having a small pure heavy-element core in Jupiter leads to the conclusion that such a compact core exists, this will usefully constrain Jupiter's origin and evolution, or alternatively, would require modifications in formation and evolution models.

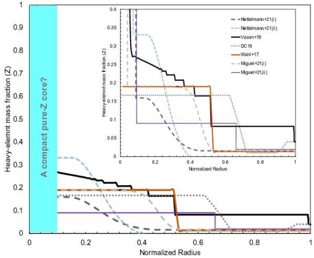


Figure 2. The heavy-element mass fr Jupiter as a function of normalizec The small panel zooms in to the beyond the core. The various correspond to different published m indicated in the figure. In some mo innermost ~1% by mass (~10% by ra the planet is assumed to be a pure element region, i.e., a compact co others can fit the gravity data without pure heavy-element core. In all the mo heavy-element mass fraction in the p deep interior is moderate.

Credit: Militzer et al. (2022), arXiv:2202.10041

- Juno measures J_2, J_4, J_6, J_8 to unprecedented precision
- Gravity inversions: heavy elements spread over inner 30–50% of radius
- **No compact core** reproduces the data
- Smooth $Z(r)$ gradient
- Challenges standard core accretion (Lecture 2)
- Explanations: convective erosion; giant-impact-driven mixing

Ganymede: The Largest Moon

Ganymede (R = 2634 km, larger than Mercury) has a complex differentiated interior.

- Four layers: solid iron inner core, liquid iron outer core, silicate mantle, ice shell/ocean
- Subsurface ocean: ~100 km deep, beneath an ice shell ~150 km thick
- Ocean sandwiched between ice layers of different phases (ice I above, ice III/V/VI below)
- Confirmed by Hubble UV observations of aurora (Saur et al. 2015)
- Has its own intrinsic magnetic field (from liquid iron core dynamo)
- JUICE will dedicate most of its mission to mapping Ganymede

Source: Anderson et al. (1996); Saur et al. (2015)

Superionic Ice: Ice Giants' Exotic Interior

At the temperatures and pressures inside Uranus and Neptune, water ice adopts an extraordinary phase.

- **Superionic ice:** oxygen atoms form a rigid crystalline lattice; hydrogen ions diffuse like a liquid
- Occurs at $P > 100$ GPa, $T > 2000$ K
- First confirmed experimentally by Millot et al. (2019) using laser-driven shocks
- Predicted to constitute most of the "ice" layer in Uranus and Neptune
- Electrical conductivity of superionic ice is high ⇒ supports dynamo generation
- Explains the tilted, multipolar magnetic fields of the ice giants

Source: Millot et al. (2019), *Nature*; Ravasio et al. (2021)

PREM: Seismic Velocities

The Preliminary Reference Earth Model gives V_p and V_s vs. depth.

- V_p (P-wave): 6–14 km/s from crust to inner core
- V_s (S-wave): 3–7 km/s in solid layers; **zero** in the outer core
- Discontinuities at the Moho (crust–mantle), 410 km, 660 km, and CMB
- Outer core S-wave gap is the key evidence for a liquid outer core
- Inner core S-wave velocity is low (~3.5 km/s) but nonzero ⇒ solid
- Combined with density, constrains composition and phase via Birch's law

Source: Dziewonski & Anderson (1981), *Phys. Earth Planet. Inter.*

Europa Clipper: Characterising an Ocean World

- Launched October 2024, arrival at Jupiter system 2030
- Dozens of close Europa flybys (not an orbiter)
- Key instruments:
  - Ice-penetrating radar (REASON): ice shell thickness
  - Magnetometer: ocean salinity and depth
  - Gravity science: internal mass distribution
  - Mass spectrometer: plume/surface composition

Europa Clipper will provide the most detailed constraints yet on the internal structure and habitability of an ocean world.

Summary

1. Seismology, gravity, and C/MR^2 are the primary tools for probing interiors
2. $C/MR^2 = 2/5$ for uniform sphere; lower values indicate differentiation
3. Hydrostatic equilibrium + equations of state determine interior profiles
4. Earth: crust, mantle (with phase transitions at 410, 660 km), liquid outer core, solid inner core
5. Mantle is solid but convects ($\mathrm{Ra} \sim 10^7$–10^8), driving plate tectonics
6. Giant planets: molecular → metallic H, dilute cores (Jupiter), helium rain (Saturn)
7. Ice giants: superionic water, tilted non-dipolar magnetic fields
8. Ocean worlds (Europa, Enceladus, Titan) expand the concept of habitability

Questions?

Next: **Lecture 9. Rocky Planets: Earth & Venus**

Lecture notes: `formingworlds.github.io/IntroductionToPlanetaryScience`